Supplement to "The temporal geometry of bandits with arriving arms"

This supplement uses the deterministic arrival model and notation of the main article unless a result explicitly introduces a different experiment. All main risk bounds concern expected pseudo-regret at deterministic endpoints. Constants are universal unless a dependence is stated. The proofs use the original independent potential reward streams; they do not condition on realized old observations and then regard those observations as fresh.

Contents

S1 Proof of the spectral minimax theorem **2**

S1.1 Concentration on potential sample streams . . . 2
S1.2 The full-history segment bound . . . 3
S1.3 Charging clock segments to the spectrum . . . 4
S1.4 A compatible family for the lower bound . . . 5
S1.5 A lower bound with constant probability . . . 7

S2 Capacity, scalar summaries and stability **8**

S2.1 Feasible fractional allocations . . . 8
S2.2 Hull and dual calculations . . . 8
S2.3 The lifetime fixed point . . . 9
S2.4 Exact capacity sensitivity . . . 10
S2.5 The learning boundary and rate examples . . . 11

S3 Instance bounds and further consequences **12**

S3.1 A detailed gap-dependent inequality . . . 12
S3.2 The cost of erasing evidence . . . 13
S3.3 Safe clipping and inserted observations . . . 14
S3.4 Time-uniform and almost-sure consequences . . . 14
S3.5 Conditioning and future information . . . 15

S4 A cut representation and diminishing marginal capacity **15**

S5 The sharp cost of requiring integer allocations **17**

S6 Why deterministic-schedule lower bounds cannot be averaged **19**

S7 A positive characterization for fixed batch templates **21**

S1 Proof of the spectral minimax theorem

S1.1 Concentration on potential sample streams

If $X \in [0,1]$ has mean μ, then

$$\mathbb{E}e^{\lambda(\mu - X)} \le e^{\lambda^2/8}, \qquad \lambda \in \mathbb{R}. \tag{S1}$$

For completeness, the log moment generating function has value and derivative zero at the origin. Its second derivative is the variance under an exponentially tilted law, at most $1/4$ because its support remains in an interval of length one. Integrating twice proves (S1).

For $S_n = \sum_{r=1}^{n}(\mu - X_r)$, the process $e^{\lambda S_n - n\lambda^2/8}$ is a nonnegative supermartingale. Stopping it at the first threshold crossing before a deterministic integer N shows that the probability of ever exceeding c by time N is at most $1/c$. Monotone convergence of the crossing events gives the same inequality over all n. Taking $\lambda = 4x$ or $\lambda = 4y/N$, respectively, gives

$$\mathbb{P}\{\exists n > v : S_n/n \ge x\} \le e^{-2vx^2}, \quad v, x > 0, \tag{S2}$$

$$\mathbb{P}\{\max_{n \le N} S_n \ge y\} \le e^{-2y^2/N}, \quad y > 0. \tag{S3}$$

The one-prefix upper-tail inequality is also $\mathbb{P}(\widehat{\mu}_n - \mu \ge x) \le e^{-2nx^2}$.

Lemma S1. *The random variable in* (10) *satisfies, for* $v, x, z > 0$,

$$\mathbb{E}Z(v) \le 5/\sqrt{v}, \qquad \mathbb{P}\{Z(v) > x\} \le \min\{1, 5/(vx^2)\}, \qquad \mathbb{E}[Z(v)\mathbf{1}\{Z(v) \ge z\}] \le 10/(vz).$$

Proof. Use (S2) for $n > v$. For $n \le v$, let $m_k = v2^{-(k+1)}$ and partition into $m_k < n \le 2m_k$, $k \ge 0$. If a prefix in this interval makes $Z(v) > x$, then

$$S_n > m_k x + \sqrt{m_k k \log 2}.$$

For each nonempty integer interval, (S3) with $N = \lfloor 2m_k \rfloor$ bounds its probability by $2^{-k}e^{-vx^2/2^{k+1}}$. Thus

$$\mathbb{P}\{Z(v) > x\} \le e^{-2vx^2} + \sum_{k \ge 0} 2^{-k} e^{-vx^2/2^{k+1}}. \tag{S4}$$

Integrating these nonnegative terms gives

$$\mathbb{E}Z(v) \le \frac{\sqrt{\pi}}{2\sqrt{2v}} + \frac{\sqrt{\pi}}{\sqrt{2v}} \sum_{k \ge 0} 2^{-k/2} < \frac{5}{\sqrt{v}}.$$

For the polynomial tail, put $u_k = vx^2/2^{k+1}$. Since $u_k e^{-u_k} \le 2\int_{u_k/2}^{u_k} e^{-u}\,du$ and these integration intervals are disjoint, the sum in (S4) is at most $4/(vx^2)$. Its first term is at most $1/(vx^2)$. Finally,

$$\mathbb{E}[Z\mathbf{1}\{Z \ge z\}] = z\mathbb{P}(Z \ge z) + \int_z^\infty \mathbb{P}(Z > x)\,dx \le \frac{10}{vz}.$$

Non-strict tail inequalities follow by limits. All suprema are countable, and all interchanges use nonnegative integrands. □

Lemma S2 (Distinct counts)**.** *Consider at most h selected pairs (i,n) from k fixed potential streams, where $n \geq 1$ and each pair is selected at most once. Selection may depend on the entire adaptive experiment. Then*

$$\mathbb{E} \sum_{\text{selected } (i,n)} (\widehat{\mu}_{i,n} - \mu_i)_+ \leq \sqrt{2kh}.$$

Also, for $v > 0$,

$$\sum_{n \geq 1} \sqrt{[\log(v/n)]_+ / n} \leq \sqrt{2\pi v}.$$

Proof. For every $\eta > 0$, the first sum is bounded pathwise by

$$\eta h + \sum_{i=1}^{k} \sum_{n \geq 1} (\widehat{\mu}_{i,n} - \mu_i - \eta)_+.$$

The expectation of the double sum for each stream is at most

$$\int_\eta^\infty \sum_{n \geq 1} e^{-2nx^2} \, dx = \int_\eta^\infty \frac{dx}{e^{2x^2} - 1} \leq \frac{1}{2\eta}.$$

Minimize at $\eta = \sqrt{k/(2h)}$; the case of no selections is immediate. For the second bound the summand is a decreasing function on $(0, v)$, so its sum is at most

$$\int_0^v \sqrt{\log(v/x)/x} \, dx = \sqrt{v} \int_0^\infty u^{1/2} e^{-u/2} \, du = \sqrt{2\pi v}.$$

This also covers $v < 1$, when the sum is zero. □

S1.2 The full-history segment bound

Proof of Lemma 1. If $d > h$, use $R_{\text{segment}} \leq h$. Otherwise at most $k \leq d$ arms are available during the prefix. Genuine first pulls cost at most $d \leq \sqrt{dh}$. At every remaining selection the best available arm has been observed, because an unobserved best arm would force the selection of an unobserved arm. Apply (12). For each local time s, a best-arm label is fixed by the deterministic schedule and the fixed reward means, so Lemma S1 gives a total expected underestimation cost at most

$$5 \sum_{s=1}^{h} \sqrt{d/s} \leq 10\sqrt{dh}.$$

Lifetime count pairs do not repeat. Lemma S2 bounds the empirical overestimation sum by $\sqrt{2kh} \leq \sqrt{2dh}$. For the bonus replace s by h, extend each arm's counts to all positive integers, and apply the same lemma to obtain $k\sqrt{2\pi h/d} \leq \sqrt{2\pi dh}$. Summing gives (13). The samples can include arbitrary observations gathered before the segment. The proof has used only unconditional prefix inequalities on their original streams. □

S1.3 Charging clock segments to the spectrum

Write $f_b(k) = \min\{b, \sqrt{kb}\}$. We will use

$$f_b\left(\sum_r k_r\right) \le \sum_r f_b(k_r).$$

If any right-hand term is saturated the inequality follows at once; otherwise it follows from subadditivity of the square root.

Lemma S3. *Deleting competing arms and delaying the remaining arms, with the endpoint fixed, produces a spectrum $\mathscr{S}'_T$ satisfying $\mathscr{S}'_T \le 2\mathscr{S}_T$.*

Proof. Arms in original group m can only move to groups $\ell \le m$. If $k_{m,\ell}$ counts them, their new total contribution is at most both

$$\sum_{\ell \le m} b_\ell < 2b_m, \qquad \sum_{\ell \le m} \sqrt{k_{m,\ell} b_\ell} \le \sqrt{2k_m b_m}.$$

It is therefore at most $2f_{b_m}(k_m)$. Subadditivity separates the contributions of different original groups, and summing proves the claim. □

Lemma S4. *If at least m competitors each have lifetime at least H, then $\mathscr{S}_T \ge \min\{H, \sqrt{mH}\}$.*

Proof. Every group containing those arms has upper endpoint at least H. Its cost is at least f_H applied to the number of these arms in the group. Subadditivity of f_H then gives the assertion. □

Upper bound in Theorem 1. Before the first competitor arrives, regret is zero. Index subsequent clock segments by $j = 0, \dots, J$, with start σ_j, length h_j, initial competitor count $c_j = K_{\sigma_j} - 1$, capacity $d_j = 2c_j$, and remaining lifetime $H_j = T - \sigma_j + 1$. Then $c_{j+1} \ge 2c_j$.

If some $c_j \ge H_j$, let j_* be the first such index. The regret of this segment and all later rounds is at most $H_{j_*} \le \mathscr{S}_T$ by Lemma S4. If there is no such index, there is no tail of this type. For the preceding segments set $c_{-1} = 0$ and let E_j consist of the competitors arriving between consecutive segment starts, including all initial competitors for $j = 0$. Then

$$m_j = |E_j| = c_j - c_{j-1} \ge c_j/2.$$

Delete other arms and delay every arm in E_j to σ_j. Its new spectrum obeys $\mathscr{S}'_T \le 2\mathscr{S}_T$.

Group the retained segment starts by the dyadic scale of H_j. For a nonempty scale ℓ, let j_{last} be its last segment. Geometric growth implies

$$\sum_{j:\, H_j \in (b_\ell/2, b_\ell]} \sqrt{c_j H_j} \le \frac{\sqrt{c_{j_{\text{last}}} b_\ell}}{1 - 2^{-1/2}}.$$

The number k'_ℓ of delayed competitors in this group satisfies

$$c_{j_{\text{last}}}/2 \le k'_\ell \le c_{j_{\text{last}}} < H_{j_{\text{last}}} \le b_\ell.$$

The group is unsaturated, and the preceding sum is at most $\sqrt{2} f_{b_\ell}(k'_\ell)/(1-2^{-1/2})$. Summing and using $h_j \le H_j$ gives

$$\sum_{j<j_*} \sqrt{c_j h_j} \le \frac{2\sqrt{2}}{1-2^{-1/2}} \mathscr{S}_T,$$

where the sum includes every segment when no j_* exists. Lemma 1 now gives

$$\mathbb{E} R_T \le \left(1 + \frac{60}{1-2^{-1/2}}\right) \mathscr{S}_T < 210 \mathscr{S}_T.$$

Together with $R_T \le T$, this proves the stated, slightly looser constant. The endpoint-dependent grouping is used only in the proof. □

S1.4 A compatible family for the lower bound

The likelihood argument used below is standard in bandit lower bounds; see Kaufmann et al. (2016). We give its precise form here. If two Bernoulli environments differ only on arm i, then their history laws through a deterministic round u satisfy

$$D(\mathbb{P}_0^u \| \mathbb{P}_i^u) = \mathbb{E}_0 N_i(u)\, \mathrm{kl}(p_i, q_i). \tag{S5}$$

To prove it, factor the conditional likelihood at each action and reward. The action kernels are the same under the two environments and cancel; only observed rewards from arm i remain. Policy randomization can be included in the history and contributes zero divergence. For $0 \le Y \le h$,

$$|\mathbb{E}_0 Y - \mathbb{E}_i Y| \le h\, \mathrm{TV}(\mathbb{P}_0^u, \mathbb{P}_i^u) \le h\sqrt{D(\mathbb{P}_0^u \| \mathbb{P}_i^u)/2}. \tag{S6}$$

The second inequality follows by binary data processing and $\mathrm{kl}(p,q) \ge 2(p-q)^2$. The latter follows by differentiating twice in p and using $1/\{p(1-p)\} \ge 4$. We also use $\mathrm{kl}(p,q) \le (p-q)^2/\{q(1-q)\}$, a consequence of $\log x \le x - 1$.

A two-environment component. Suppose $M \ge 1$, and choose a competitor with maximum lifetime $H_{\max}$. Set all other means to $1/2$, and set this competitor's mean to $1/2 \pm \epsilon$, where $\epsilon = 1/(32\sqrt{H_{\max}})$. All rewards are Bernoulli. The divergence between the two complete history laws is at most $32 H_{\max} \epsilon^2$, so their total variation is at most $1/8$. If N counts this arm's pulls after it arrives, the equal-prior Bayes risk is at least

$$\frac{\epsilon}{2}\{\mathbb{E}_- N + H_{\max} - \mathbb{E}_+ N\} \ge \frac{\epsilon H_{\max}}{2}(1 - 1/8) \ge \frac{\sqrt{H_{\max}}}{128}. \tag{S7}$$

Let S_{rest} be the spectral mass in groups with $k_\ell = 1$ or $\ell \le 1$. A geometric sum gives

$$S_{\mathrm{rest}} \le \frac{\sqrt{2H_{\max}}}{1-2^{-1/2}} + 3 \le 8\sqrt{H_{\max}}.$$

Thus the minimax risk is at least $S_{\mathrm{rest}}/1024$.

Selecting compatible scales. For the other groups, $\ell \geq 2$ and $k_\ell \geq 2$. Let their total cost be S_{high}, and put

$$m_\ell = k_\ell - 1, \qquad v_\ell = \min\{1, \sqrt{m_\ell / b_\ell}\}.$$

Then $b_\ell v_\ell \geq s_{\ell,T}/\sqrt{2}$. Choose either the even or the odd groups, whichever has greater $\sum b_\ell v_\ell$. Scan the chosen groups from larger to smaller lifetimes. Retain the first, then retain a group only if its v_ℓ exceeds twice the most recently retained value. Call the retained set $\mathcal{J}$.

Within a fixed parity, b_ℓ decreases by a factor of at least four. Between two retained groups, omitted values have $v \leq 2v_{\text{last}}$, and their b values sum to at most $b_{\text{last}}/3$. Their total mass is therefore at most two thirds of the preceding retained mass. It follows that

$$\sum_{\ell \in \mathcal{J}} b_\ell v_\ell \geq \frac{3}{10\sqrt{2}} S_{\text{high}}, \qquad \sum_{\ell \in \mathcal{J}} v_\ell \leq 2. \tag{S8}$$

The second inequality holds because retained values more than double in chronological order and none exceeds one.

One product prior. For each retained group let

$$\epsilon_\ell = v_\ell/32, \qquad \theta_\ell = \frac{1}{2} + \sum_{r \in \mathcal{J}: r > \ell} \epsilon_r.$$

Give one actual arm in the group mean θ_ℓ. This is its reference arm. Among its m_ℓ remaining candidates, put probability $1/2$ on the null configuration, in which all means equal $\theta_\ell - \epsilon_\ell$. Give each candidate i probability $1/(2m_\ell)$ of being the unique spike with mean $\theta_\ell + \epsilon_\ell$, leaving all other candidates at the null mean. Choose these latent configurations independently across groups. The anchor has mean $1/2$, and all unused arms have mean $1/4$. All rewards are Bernoulli.

Equation (S8) gives $\sum \epsilon_\ell \leq 1/16$, so all parameters remain in a fixed interior subinterval of $[0, 1]$. Reference levels depend only on deterministic gap sizes, not on previous latent configurations. Observing later reference arms therefore does not reveal earlier configurations.

Charge group ℓ only on the interval

$$Q_\ell = \{T - b_\ell/2 + 1, \ldots, T - b_\ell/4\}, \qquad h_\ell = b_\ell/4. \tag{S9}$$

These are integer intervals. Every arm in group ℓ has arrived before Q_ℓ. The next retained group has not arrived by its end, and the intervals are disjoint. They remain within $[1, T]$ even when $b_\ell > T$, since a nonempty group implies $T > b_\ell/2$. Every earlier group's largest possible mean is at most θ_ℓ. Consequently, on Q_ℓ the reference is optimal under the null and candidate i is optimal under spike i.

Testing with the complete observed history. Fix all latent configurations other than the one in group ℓ. Under its null write

$$A = \sum_{i=1}^{m_\ell} \mathbb{E}_0 N_i(Q_\ell), \qquad D_i = \mathrm{kl}(\theta_\ell - \epsilon_\ell, \theta_\ell + \epsilon_\ell) \mathbb{E}_0 N_i(\max Q_\ell).$$

Here $N_i(Q_\ell)$ counts pulls in the window, whereas $N_i(\max Q_\ell)$ counts all pulls through its end. The latter includes information acquired before the window. At most b_ℓ physical slots are available between the group's first possible arrival and this endpoint. The KL bound and Cauchy–Schwarz therefore give

$$\frac{1}{m_\ell}\sum_i \sqrt{D_i/2} \le \sqrt{\frac{32\epsilon_\ell^2 b_\ell}{2m_\ell}} \le \frac{1}{8}. \tag{S10}$$

The null-window regret is at least $\epsilon_\ell A$, and under spike i it is at least $\epsilon_\ell\{h_\ell - \mathbb{E}_i N_i(Q_\ell)\}$. Equation (S6) gives conditional Bayes risk at least

$$\begin{aligned}\frac{\epsilon_\ell}{2}A + \frac{\epsilon_\ell}{2m_\ell}\sum_i\{h_\ell - \mathbb{E}_i N_i(Q_\ell)\} &\ge \frac{\epsilon_\ell}{2}\left\{h_\ell + (1-1/m_\ell)A - h_\ell\frac{1}{m_\ell}\sum_i\sqrt{D_i/2}\right\}\\ &\ge \frac{\epsilon_\ell h_\ell}{4} = \frac{b_\ell v_\ell}{512}.\end{aligned}$$

This holds for every configuration of the other groups. Average over those configurations and sum over the disjoint windows. The resulting lower bound concerns one product prior, not a sum of unrelated worst-case environments. By (S8),

$$\mathfrak{R}_T(\tau) \ge \frac{3}{5120\sqrt{2}} S_{\text{high}} \ge S_{\text{high}}/2500.$$

Taking the maximum with (S7) and the rest-mass bound gives $\mathfrak{R}_T(\tau) \ge \mathscr{S}_T/10000$. For $M = 0$ both sides vanish. This proves the lower bound in Theorem 1.

The anchor law remains Bernoulli(1/2) in both constructions. The lower bound therefore also holds if the learner is told the anchor's complete reward law in advance.

S1.5 A lower bound with constant probability

Proposition S5. *For every deterministic schedule, endpoint and policy, there is a fixed Bernoulli environment such that*

$$\mathbb{P}\{R_T \ge \mathscr{S}_T/20000\} \ge 1/8.$$

The environment may depend on the policy, schedule and endpoint.

Proof. If $S_{\text{high}} \ge \mathscr{S}_T/2$, use the product prior above. On each window define $Y_\ell = \epsilon_\ell \sum_i N_i(Q_\ell)$ under the null and $Y_\ell = \epsilon_\ell\{h_\ell - N_i(Q_\ell)\}$ under spike i. Then, pathwise,

$$0 \le Y := \sum_\ell Y_\ell \le R_T, \qquad Y \le A := \sum_\ell \epsilon_\ell h_\ell, \qquad \mathbb{E}Y \ge A/4.$$

Since $\mathbb{E}Y \le A/8 + A\mathbb{P}(Y \ge A/8)$, we obtain $\mathbb{P}(Y \ge A/8) \ge 1/8$. Equation (S8) yields

$$A/8 = \frac{1}{1024}\sum_{\ell\in\mathcal{J}} b_\ell v_\ell \ge \frac{3\mathscr{S}_T}{20480\sqrt{2}} \ge \frac{\mathscr{S}_T}{20000}.$$

If $S_{\text{rest}} \ge \mathscr{S}_T/2$, use the two-environment prior. Its proxy loss is ϵN under the inferior alternative and $\epsilon(H_{\max} - N)$ under the superior alternative. It lies in $[0, A]$ with $A = \sqrt{H_{\max}}/32$, has mean at least $A/4$, and satisfies $A/8 \ge \mathscr{S}_T/4096$. The same probability argument applies. In either case a

finite-prior average probability of at least $1/8$ implies that at least one fixed environment attains that probability. The zero-competitor case is immediate. □

Define $r_\delta = \inf_\pi \inf\{r \geq 0 : \sup_P \mathbb{P}_P(R_T > r) \leq \delta\}$. For $0 < \delta < 1/8$, the proposition and Markov's inequality give

$$\mathscr{S}_T/20000 \leq r_\delta \leq \min\{T, 300\mathscr{S}_T/\delta\}.$$

This characterizes fixed confidence levels up to constants. It does not give sharp dependence as δ tends to zero.

S2 Capacity, scalar summaries and stability

S2.1 Feasible fractional allocations

The necessity of (5) follows by counting the available slots of arms with lifetime at most h. For sufficiency, order the arms increasingly by lifetime. At the step for arm i, all previously assigned mass lies in the suffix of length H_i, because its arms have shorter lifetimes. The total previous mass plus n_i is at most H_i by the corresponding constraint. There is therefore enough unused capacity in that suffix to assign n_i. Induction constructs the fractional schedule. The same argument works with whole slots when all n_i are integers.

Proof of Theorem 2. For any feasible n, let $a_\ell = \sum_{i \in G_\ell} n_i$. This is at most b_ℓ, either by the suffix constraint or, if $b_\ell > T$, by the total-horizon constraint. Since $\phi(x) \leq x$ and $\phi(x) \leq \sqrt{x}$, Cauchy–Schwarz gives

$$\sum_{i \in G_\ell} \phi(n_i) \leq \min\{a_\ell, \sqrt{k_\ell a_\ell}\} \leq s_{\ell,T}.$$

Summing proves $\mathscr{C}_T \leq \mathscr{S}_T$.

For the reverse comparison set $n_i = b_\ell/(4k_\ell)$ for $i \in G_\ell$, for every nonempty group. Fix an integer suffix h. All arms with $H_i \leq h$ belong to groups with upper endpoints at most $b_{\max} < 2h$, including the group $b_0 = 1$. Counting the entire groups only increases their total demand, and

$$\sum_{i : H_i \leq h} n_i \leq \frac{1}{4} \sum_{b_\ell \leq b_{\max}} b_\ell < \frac{b_{\max}}{2} < h.$$

The allocation is feasible. Its contribution in group ℓ is

$$k_\ell \phi\left(\frac{b_\ell}{4k_\ell}\right) = \min\{b_\ell/4, \sqrt{k_\ell b_\ell}/2\} \geq s_{\ell,T}/4.$$

The minimax comparisons follow directly from Theorem 1. Empty competitor sets give zero throughout. □

S2.2 Hull and dual calculations

Proof of Proposition 2. For a direct construction, from the current vertex $(r_{b-1}, H_{(r_{b-1})})$ choose the largest $r > r_{b-1}$ minimizing

$$\frac{H_{(r)} - H_{(r_{b-1})}}{r - r_{b-1}}.$$

The first slope is positive. If the next slope were smaller, the slope to its endpoint from the previous vertex would be smaller too, contrary to the choice of the previous minimum. The choice of the largest minimizing index removes redundant collinear vertices. Thus the slopes are positive and nondecreasing. Minimality also places each chord below all intermediate points. The allocation in (20) is feasible, with equality in its prefix constraint at every vertex.

Take a supergradient p_b of ϕ at q_b as specified after Proposition 2. At the corner $q_b = 1$, the value one is a valid supergradient. These supergradients are nonincreasing. Put $p_{B+1} = 0$, $\lambda_{r_b} = p_b - p_{b+1} \geq 0$, and zero elsewhere. If i belongs to block b, then $\sum_{r \geq i} \lambda_r = p_b$. Concavity gives, for every feasible n,

$$\begin{aligned}\sum_i \{\phi(n_i) - \phi(n_i^*)\} &\leq \sum_i p_i(n_i - n_i^*) \\ &= \sum_r \lambda_r \left(\sum_{i \leq r} n_i - \sum_{i \leq r} n_i^* \right) \\ &= \sum_r \lambda_r \left(\sum_{i \leq r} n_i - H_{(r)} \right) \leq 0.\end{aligned}$$

The third line uses tightness only where the multiplier is nonzero. This proves optimality without strict concavity or uniqueness.

Direct one-dimensional maximization on $[0, 1]$ and $[1, \infty)$ gives $g(c) = \sup_{x \geq 0}\{\phi(x) - cx\}$ with the formula in the main article. For any $\lambda \geq 0$, set $c_i = \sum_{r \geq i} \lambda_r$. The inequality $\phi(n_i) \leq c_i n_i + g(c_i)$ and the prefix constraints give the upper dual objective in (22). For the multipliers just constructed, each c_i is a supergradient at n_i^*, so equality holds in the conjugate inequality. Complementary tightness also holds. This supplies an attaining dual certificate and proves exact equality.

After sorting, the usual monotone stack removes a middle point whenever the last two slopes cease to increase. Each point enters and leaves the stack at most once. This proves the stated arithmetic operation count. □

S2.3 The lifetime fixed point

Proof of Theorem 3. When $M \geq 1$, $Q_T \geq 1$. Put $r = \lceil Q_T \rceil$. If $r > T$, then $\Gamma_T \leq T < r$. Otherwise every arm with $H_i > r$ belongs to a group with $b_\ell > Q_T$. This group cannot be saturated, since then $s_{\ell,T} = b_\ell > Q_T$. Therefore

$$V_T(r) \leq \sum_{\ell : b_\ell > Q_T} k_\ell b_\ell = \sum_{\ell : b_\ell > Q_T} s_{\ell,T}^2 \leq Q_T^2 \leq r^2.$$

This proves $\Gamma_T \leq \lceil Q_T \rceil \leq 2Q_T$.

Conversely, set $r = \Gamma_T$ and $m = \lceil \log_2 r \rceil$. The small groups satisfy

$$\sum_{\ell \leq m} s_{\ell,T}^2 \leq \sum_{\ell \leq m} 4^\ell < \frac{16}{3} r^2.$$

All lifetimes in groups $\ell > m$ exceed r, and $b_\ell < 2H_i$ there. Hence

$$\sum_{\ell>m} s_{\ell,T}^2 \le \sum_{\ell>m} k_\ell b_\ell \le 2V_T(r) \le 2r^2.$$

The total is less than $9r^2$, proving $Q_T \le 3\Gamma_T$. Cauchy–Schwarz and nonnegativity give $1 \le \mathscr{S}_T^2/Q_T^2 \le L+1$. Combining with Theorem 1 gives the effective-scale statement.

For the final refinement, the small-group sum is less than $4r$, while at most $(L-m)_+$ large groups have squared sum at most $2r^2$. Thus

$$\mathscr{S}_T \le 4r + r\sqrt{2(L-m)_+} \le Cr\sqrt{1+\log(T/r)}.$$

The comparison uses a universal constant, including the case $r = T$. □

For the family in (19), group j has $4^j H_j = T$ and $4^j \le H_j$, so its cost is $\sqrt{T}$. Summing the costs and their squares yields both equalities in that display. Equation (17) then yields $\Gamma_T \asymp \sqrt{mT}$. In the comparison schedule, the $m+1$ old competitors contribute $\sqrt{(m+1)T}$ and the last-round competitors contribute one. Its squared spectral norm has the same order as before. This verifies both the separation and the assertion about (K_T, Γ_T).

For a general sufficiently large endpoint T, choose the largest $16^m \le T$ and place the same experiment in the last 16^m rounds, keeping only the anchor before that. The competitor lifetimes do not change and $m \asymp \log T$. Together with the upper comparison in Theorem 3, this proves that the largest possible order of $\mathscr{S}_T/\Gamma_T$ is $\sqrt{\log(eT)}$.

S2.4 Exact capacity sensitivity

Proposition S6. *Adding any finite batch whose lifetimes are at most h increases capacity by an amount in $[0, h]$. The jitter inequality (24) and one-step growth inequality (25) also hold. With at least one competitor,*

$$\sqrt{H_{\max}} \le \mathscr{C}_T \le H_{\max} \le T.$$

Proof. Adding arms enlarges the feasible set. From a maximizing allocation in the enlarged problem, delete the new arms' mass. Their total mass is at most h, and their total objective is no larger because $\phi(x) \le x$. The remaining allocation is feasible for the old problem, proving the batch bound.

For jitter, take a feasible allocation n under H' and order its labels by increasing H. The first r labels all have $H'_i \le H_{(r)} + d$, so their total allocated mass is at most $H_{(r)} + d$. If this exceeds the horizon, the total-horizon bound still implies the inequality. Visit the prefixes in order and delete just enough mass to meet each original constraint. Earlier upper constraints cannot be hurt by deletion. The cumulative deleted mass after prefix r can be chosen as

$$D_r = \max_{q \le r} \left(\sum_{i \le q} n_{(i)} - H_{(q)} \right)_+ \le d.$$

At each step the mass to delete is available in that prefix, because the target bound is nonnegative. The total objective falls by at most d, since ϕ is 1-Lipschitz. This proves $\mathscr{C}(H') \le \mathscr{C}(H) + d$. Interchange the lifetime vectors for the reverse inequality.

To advance the physical endpoint by one, keep the old physical allocation to prove monotonicity. In an optimal allocation for the new endpoint, delete the last time slot. All mass assigned to newly arrived arms disappears; the remaining assignment is valid by the old endpoint. At most one unit of mass was deleted, so the objective decreases by at most one. This proves (25) even with a large new batch.

Finally, all competing arms are confined to the last $H_{\max}$ slots, so their total objective is at most their total mass, at most $H_{\max}$. Assigning all $H_{\max}$ slots to a longest-lived competitor gives objective $\phi(H_{\max}) = \sqrt{H_{\max}}$. □

S2.5 The learning boundary and rate examples

Proof of Theorem 4. Since $\sum_{\ell=0}^{L} b_\ell < 4T$, Cauchy–Schwarz gives

$$\mathscr{S}_T \le \sum_{\ell} \sqrt{k_\ell b_\ell} \le 2\sqrt{(K_T - 1)T}.$$

If $K_T = o(T)$, Theorem 1 gives uniform sublinear regret for the same MC-MOSS rule. Conversely, suppose $K_{t_j} \ge \epsilon t_j$ for some $\epsilon > 0$ along $t_j \to \infty$. At $T_j = 2t_j$, at least $\epsilon t_j - 1$ competitors have lifetime at least $t_j + 1$. Lemma S4 gives $\mathscr{S}_{T_j} \ge c_\epsilon T_j$ for all large j. The minimax lower bound prevents any policy from having uniform sublinear regret. Equivalence with capacity follows from Theorem 2.

For the probability formulation, Markov's inequality gives one direction. For the other, use $0 \le R_T/T \le 1$ to write, for any $\epsilon > 0$,

$$\sup_P \mathbb{E}_P(R_T/T) \le \epsilon + \sup_P \mathbb{P}_P(R_T/T > \epsilon).$$

Let T tend to infinity, then ϵ tend to zero. □

If M competitors share a lifetime H, Jensen's inequality gives

$$\mathscr{C}_T = M\phi(H/M) = \min\{H, \sqrt{MH}\},$$

with the equal allocation attaining it. This includes a single late batch and the static model; it is interpreted as zero for $M = 0$.

If $K_{\lfloor T/2 \rfloor} - 1 \ge a(K_T - 1) > 0$ for a fixed $a \in (0, 1]$, these early competitors have lifetime at least $T/2$. Lemma S4 and the preceding general upper bound give

$$\mathfrak{R}_T(\tau) \asymp_a \min\{T, \sqrt{(K_T - 1)T}\}.$$

For each growth family listed in the main text, the two-sided growth bounds imply this early-fraction condition eventually. Substitution yields the stated orders, with constants allowed to depend on the growth parameters.

S3 Instance bounds and further consequences

S3.1 A detailed gap-dependent inequality

Let $\mathcal{B}_T$ be the arms that are strictly suboptimal at some time by T, with γ_i as in the main article. Set $L_T^0 = \log(eT)$. At time t, let

$$\gamma_t = \min_{i \in A_t : \Delta_i(t) > 0} \Delta_i(t),$$

and set $\gamma_t = +\infty$ when all available means are equal. If the current clock segment has capacity d_j and local time s, define

$$\mathcal{U}_T = \sum_{t \le T : \gamma_t < \infty} \min\left\{1, 15\sqrt{d_j/s}, \frac{90 d_j}{s\gamma_t}\right\}. \tag{S11}$$

There are no summands before a competitor is present.

Proposition S7. *MC-MOSS satisfies*

$$\mathbb{E}R_T \le \sum_{i \in \mathcal{B}_T} \left\{1 + \frac{9 + 18L_T^0}{\gamma_i}\right\} + \mathcal{U}_T. \tag{S12}$$

Proof. Genuine first observations of strictly suboptimal arms cost at most $|\mathcal{B}_T|$. At each other positive-regret selection, write (12) as $\Delta \le Z + e + b$. Assign the selection to the first of the following events that holds: $Z \ge \Delta/3$, $e \ge \Delta/3$, $b \ge \Delta/3$. At least one holds.

On the underestimation event, regret is bounded by $\min\{1, 3Z\mathbf{1}\{Z \ge \gamma_t/3\}\}$. The expectation bounds in Lemma S1, the truncated expectation bound there, and the trivial bound one give the summand of (S11). The best-arm label used in Z is deterministic once the environment and schedule are fixed.

For the empirical overestimation event on arm i, we have $\Delta \le 3e$ and $e \ge \gamma_i/3$. For any $a > 0$, the one-prefix Hoeffding bound gives

$$\begin{aligned}\sum_{n \ge 1} \mathbb{E}[e_n \mathbf{1}\{e_n \ge a\}] &\le \frac{a}{e^{2a^2} - 1} + \int_a^\infty \frac{dx}{e^{2x^2} - 1} \\ &\le \frac{1}{2a} + \frac{1}{2a} = \frac{1}{a}.\end{aligned}$$

The actual count pairs do not repeat, so taking $a = \gamma_i/3$ and multiplying by three bounds this arm's contribution by $9/\gamma_i$.

For the bonus event, $b \le \sqrt{L_T^0/n}$ and $b \ge \gamma_i/3$ imply $n \le 9L_T^0/\gamma_i^2$. Extending to all integers in that range and using $\sum_{n \le u} n^{-1/2} \le 2\sqrt{u}$ bounds its regret by $18L_T^0/\gamma_i$. Adding the three categories proves (S12). □

Gap bounds in Proposition 1. A terminal best arm was never strictly suboptimal after its arrival, because the best available mean is nondecreasing and cannot exceed that arm's terminally optimal mean. Thus $|\mathcal{B}_T| \le K_T - 1$. Segment capacities obey $\sum_j d_j \le 4(K_T - 1)$, by their geometric growth. Since $\sum_{s \le h_j} 1/s \le \log(eT)$,

$$\mathcal{U}_T \le 360(K_T - 1)L_T^0/\gamma.$$

The other terms in (S12) sum to at most $28(K_T-1)L_T^0/\gamma$. Combine these inequalities, Theorem 1, and $R_T \le T$ to obtain (14).

For deterministic rewards, every observed empirical mean is exact. The current best index is at least B_t, so selecting an observed inferior arm implies

$$\Delta_i(t) \le \sqrt{L_T^0/N_i(t-1)}.$$

Its positive-regret pre-pull counts are at most L_T^0/γ_i^2 and do not repeat globally. Sum $\sqrt{L_T^0/n}$ over these counts and add the possible first observation. This gives (15). □

S3.2 The cost of erasing evidence

Restart separation in Proposition 1. For an integer $m \ge 1$, put $h = 2^{16m}$ and $T = mh$. Initially there are two deterministic arms of means $1/2$ and $1/4$. At time $jh+1$, for $j = 1, \dots, m-1$, add 2^{j-1} deterministic arms of mean $1/2$. The sole inferior arm has gap $1/4$ throughout. Segment j, indexed from zero, has $c_j = 2^j$ competitors, capacity $d_j = 2^{j+1}$, and length h.

For MC-MOSS, selecting the inferior arm at a positive lifetime count n implies

$$\frac{1}{4} \le \sqrt{\log_+\{s/(d_j n)\}/n} \le \sqrt{\log T/n}.$$

Its total pull count is at most $1 + 16\log T$, whence $R_T(\pi^{\mathrm{MC}}) \le 1/4 + 4\log T$.

For the cold policy, fix a segment and set $q = \lfloor \log h \rfloor$. Suppose its inferior arm is pulled fewer than q times in the entire segment. During the latter half, $s \ge h/2$, an unobserved inferior arm has infinite index. If it has $1 \le n < q$ observations, then

$$\frac{\log\{s/(d_j n)\}}{n} \ge \frac{\log\{h/(2d_j \log h)\}}{\log h} \ge \frac{1}{2}. \tag{S13}$$

Indeed, $d_j \le 2^m$ and $2d_j \log h \le \sqrt{h}$ for every $m \ge 1$. The inferior index is therefore at least $1/4 + 1/\sqrt{2} > 0.95$. Any optimal arm with at least $\lceil 16\log h \rceil$ segment observations has index at most $1/2 + 1/4 = 0.75$. As long as the inferior count stays below q, no such well-sampled optimal arm can be selected in the latter half. The total number of possible selections in that half is therefore at most

$$K_j \lceil 16\log h \rceil + q \le 273 m 2^m < h/2,$$

using $K_j = 2^j + 1 \le 2^m$, $\log h \le 16m$, and $m \le 2^{m-1}$. This contradicts its $h/2$ available rounds. Each segment thus pulls the inferior arm at least q times, giving

$$R_T(\pi^{\mathrm{cold}}) \ge \frac{m}{4} \lfloor \log h \rfloor.$$

Since $\log T = 16m \log 2 + \log m = \Theta(m)$, the asserted order separation follows.

Cold MOSS also obeys the spectral upper bound. Its local segments use fresh suffixes of independent potential streams and deterministic arrival-defined boundaries. Conditional on the past, the unused rewards have their original product law; this can be verified by conditioning on the finite observed prefix counts and values, or by the usual independent-stream stopping argument. The segment bound then holds and the identical clock charging completes the proof. This conditional

justification applies to unused rewards, not to the reused observations of MC-MOSS. □

S3.3 Safe clipping and inserted observations

For the clipped policy, an unobserved arm has index one and an observed arm has index $\min\{1, U_i(t)\}$. The current best arm's clipped index is at least $B_t - Z_{i_t^*}(s/d)$ whether or not it has been observed, since $B_t \le 1$. A selected observed arm's clipped index is at most its original $\widehat{\mu} + b$. Thus (12) continues to hold for every observed-arm selection. Genuine first observations are still charged once. All segment and gap proofs remain valid. For deterministic rewards the best clipped index is at least B_t, so the deterministic gap argument remains valid too.

In the reward-one example, after the initial observation the incumbent and all newly unobserved arms have clipped index one. Arrival-order tie breaking always chooses the incumbent, giving zero regret. Under the infinite-index rule, the single new arm at every round is selected and gives regret one. These are statements about these specified policies on one legal instance.

Proposition S8 (Predictable inserted choices)**.** *Keep the physical clock rules and update lifetime statistics after every observation. At a set of rounds $\mathcal{I}_T$ selected using information available before the action, permit any legal choice in place of MC-MOSS. Use the MC-MOSS index rule at all other rounds. Then*

$$\mathbb{E}R_T \le \min\left\{T, 300\mathscr{S}_T + \mathbb{E}\sum_{t\in\mathcal{I}_T}\Delta_{I_t}(t)\right\}.$$

Proof. For ordinary choices, index comparison is unchanged. The underestimation lemma controls every potential prefix, while the empirical and bonus sums use distinct count pairs. Inserted observations can skip counts at ordinary choices but cannot repeat them. The local proof thus applies to the ordinary sub-sequence, charged using the full physical segment length. Arrival-defined clock segments are unchanged. Add the actual regret of the inserted choices separately and use $R_T \le T$. □

The result compares the modified policy with a bound, not with the realized trajectory of an unmodified policy. It does not make an additional sampling or inference objective free of cost.

S3.4 Time-uniform and almost-sure consequences

Proposition S9. *For every fixed infinite deterministic schedule, fixed environment and $0 < \delta < 1$, MC-MOSS satisfies, with probability at least $1 - \delta$, simultaneously for all $T \ge 1$,*

$$R_T \le \min\left\{T, \frac{5000}{\delta}\mathscr{C}_T(1 + \log_2 \mathscr{C}_T)^2\right\}.$$

When $\mathscr{C}_T = 0$, the second term is defined to be zero.

Proof. If no competitor ever arrives, regret is zero. Otherwise capacity is nondecreasing, and the longest competitor's lifetime tends to infinity, so Proposition S6 implies $\mathscr{C}_T \to \infty$. For $m \ge 0$, let t_m be the last integer with $\mathscr{C}_{t_m} < 2^{m+1}$, skipping indices for which this set is empty. These are finite, schedule-determined endpoints. The capacity upper bound gives $\mathbb{E}R_{t_m} \le 1200\, 2^{m+1}$. Apply

Markov's inequality with error allowance $\delta_m = \delta/\{2(m+1)^2\}$ and sum the allowances, whose sum is less than δ. On the resulting event,

$$R_{t_m} \le 2400\, 2^{m+1}(m+1)^2/\delta$$

for every retained index. If $2^m \le \mathscr{C}_T < 2^{m+1}$, then $T \le t_m$ and nonnegativity of instantaneous regret implies $R_T \le R_{t_m}$. Since $2^{m+1} \le 2\mathscr{C}_T$ and $m+1 \le 1+\log_2 \mathscr{C}_T$, this gives the claimed constant after a small relaxation. Nonzero capacity is at least one, so its logarithm causes no ambiguity. □

This envelope is a consequence of the expectation bound and monotonicity. Its dependence on δ and on time is not claimed optimal.

If $\sum_{m\ge 0} \mathscr{S}_{2^m}/2^m < \infty$, then MC-MOSS satisfies $R_T/T \to 0$ almost surely in each fixed reward environment. For every positive rational ϵ, Markov's inequality makes $\sum_m \mathbb{P}(R_{2^m} > \epsilon 2^m)$ finite. Borel–Cantelli and a countable intersection give convergence along dyadic endpoints. For $2^m \le T < 2^{m+1}$,

$$R_T/T \le 2R_{2^{m+1}}/2^{m+1} \longrightarrow 0.$$

For example, $K_t = O(t^\alpha)$ with $\alpha < 1$ meets this sufficient condition. We do not deduce almost-sure convergence for every $K_t = o(t)$ from the expectation theorem.

S3.5 Conditioning and future information

For a schedule independent of the reward streams and the policy seed, condition on the complete schedule, apply Theorem 1, and average to obtain (28). Independence is required to preserve the potential-stream laws under this conditioning. No supremum over environments is interchanged with expectation in this argument. The random-schedule counterexample below explains why the analogous lower-bound interchange fails.

For a fixed schedule and endpoint, (4) also implies that MC-MOSS incurs at most 3×10^6 times the minimax risk of a learner given the complete schedule and endpoint. This loose constant follows directly by division, with the zero-competitor case treated separately. It compares two learners that both have unknown means, not a learner with an oracle that already knows the best arm.

S4 A cut representation and diminishing marginal capacity

Fix a horizon T and a finite universe E of candidate arms, with lifetimes $H_i \in \{1,\dots,T\}$. The initial anchor is not included in E. For $A \subseteq E$, let

$$\mathcal{N}(A) = \left\{ n \in \mathbb{R}_+^A : \sum_{i\in A: H_i \le h} n_i \le h, \quad h = 1,\dots,T \right\},$$

$$\mathscr{C}(A) = \max_{n\in\mathcal{N}(A)} \sum_{i\in A} \phi(n_i), \qquad \phi(x) = \min\{x, \sqrt{x}\}.$$

The empty-set value is zero. The following representation gives a second interpretation of capacity: it aggregates the maximum number of observations that can be assigned when the allocation to each arm is capped at a common level.

Theorem S10 (Capped allocation and submodular capacity)**.** *For $z > 0$ and $A \subseteq E$, define*

$$F_z(A) = \max_{n \in \mathcal{N}(A)} \sum_{i \in A} \min\{n_i, z\}.$$

Then

$$F_z(A) = \min_{h \in \{0,\dots,T\}} \{h + z\,|\{i \in A : H_i > h\}|\}\,. \tag{S14}$$

One feasible allocation attains these maxima simultaneously for every $z > 0$. Consequently,

$$\mathscr{C}(A) = \frac{1}{2}F_1(A) + \frac{1}{4}\int_1^\infty F_z(A) z^{-3/2}\,dz. \tag{S15}$$

Both F_z and $\mathscr{C}$ are normalized, monotone, submodular set functions. In particular, if $A \subseteq B \subseteq E$ and $j \in E \setminus B$, then

$$0 \le \mathscr{C}(B \cup \{j\}) - \mathscr{C}(B) \le \mathscr{C}(A \cup \{j\}) - \mathscr{C}(A) \le \sqrt{H_j}. \tag{S16}$$

Proof. For any feasible n and any integer $h \in \{0, \dots, T\}$,

$$\sum_{i \in A} \min\{n_i, z\} \le \sum_{i \in A : H_i \le h} n_i + z|\{i \in A : H_i > h\}| \le h + z|\{i \in A : H_i > h\}|.$$

At $h = 0$ the first sum is empty. This proves the upper bound in (S14).

We give an attaining allocation. If A is nonempty, write $m = |A|$ and sort its lifetimes as $H_{(1)} \le \cdots \le H_{(m)}$. Take the lower convex hull of $(0,0), (r, H_{(r)})$, $1 \le r \le m$, and assign each arm the slope of the hull segment covering its index. Denote this allocation by n^*. Its coordinates are positive and nondecreasing, its prefix sums lie below $H_{(r)}$, and its prefix sums equal $H_{(r)}$ at hull vertices. These facts can also be obtained directly by successively choosing the largest index minimizing the slope from the current vertex. In particular, $n^* \in \mathcal{N}(A)$.

For a given z, let r be the last index with $n^*_{(r)} \le z$, with $r = 0$ if there is no such index. Since hull slopes are constant on segments, r is a hull vertex, including the possible endpoints. With $H_{(0)} = 0$,

$$\sum_{i \in A} \min\{n^*_i, z\} = H_{(r)} + z(m - r).$$

When $0 < r < m$, this vertex is the last occurrence of its lifetime: otherwise feasibility and the positive next coordinate would give $H_{(r)} + n^*_{(r+1)} > H_{(r+1)} = H_{(r)}$. Thus, at $h = H_{(r)}$, there are exactly $m - r$ lifetimes strictly exceeding h. The displayed value equals one of the cuts in (S14). The upper bound already proved shows that it is the minimum cut and the maximum allocation value. The same n^* works for every z. The case A empty is immediate.

For all $x \ge 0$, direct integration gives

$$\phi(x) = \frac{1}{2}\min\{x, 1\} + \frac{1}{4}\int_1^\infty \min\{x, z\} z^{-3/2}\,dz.$$

Indeed, for $x \le 1$ the right side is x, while splitting the integral at $x > 1$ gives $\sqrt{x}$. Apply the upper bound for every F_z to an arbitrary feasible n, and then use n^*, which attains all of them, to obtain (S15). All exchanges are justified by nonnegativity. The integral is finite because $F_z(A) \le T$.

It remains to prove submodularity. Fix $z > 0$, and for two sets A, B choose minimizers a, b in (S14). By exchanging the names of the sets, assume $a \le b$. Write $N_D(h) = |\{i \in D : H_i > h\}|$. The following identity is nonnegative:

$$N_A(a) + N_B(b) - N_{A\cup B}(b) - N_{A\cap B}(a) = |\{i \in A \setminus B : a < H_i \le b\}| \ge 0.$$

Using b as a cut for $A \cup B$ and a as a cut for $A \cap B$ gives

$$\begin{aligned} F_z(A \cup B) + F_z(A \cap B) &\le a + b + zN_{A\cup B}(b) + zN_{A\cap B}(a) \\ &\le a + b + zN_A(a) + zN_B(b) = F_z(A) + F_z(B). \end{aligned}$$

This proves submodularity of F_z. Normalization is immediate, and monotonicity follows by extending a feasible allocation by zeros. Equation (S15), whose weights are nonnegative, proves the same properties for $\mathscr{C}$. The first two inequalities in (S16) are monotonicity and diminishing returns. For the last, compare with the empty set and note that $\mathscr{C}(\{j\}) = \phi(H_j) = \sqrt{H_j}$ because $H_j \ge 1$. □

This result concerns the deterministic capacity functional. Constant factor equivalence between $\mathscr{C}(A)$ and minimax regret does not imply submodularity of the minimax value itself. It also does not justify discarding available arms in the original decision problem: deleting an arm changes the benchmark. The result describes how the complexity of a family of decision problems changes as candidate arms are added. The general submodularity mechanism is already covered by separable concave optimization over polymatroids (He et al., 2012): here the underlying rank function is $r(D) = \max_{i\in D} H_i$, with $r(\varnothing) = 0$. The contribution of this specialization is the explicit cut integral and its interpretation for arrival schedules; submodularity itself is not claimed as a new general result.

S5 The sharp cost of requiring integer allocations

The fractional definition has an almost exact counterpart in terms of actual integer sample counts. For a fixed lifetime vector, write

$$\mathscr{C} = \max_{n\in\mathcal{N}} \sum_i \phi(n_i), \qquad \mathscr{C}_{\text{int}} = \max_{N\in\mathcal{N}\cap\mathbb{Z}_+^m} \sum_i \sqrt{N_i}.$$

Here $\mathcal{N}$ is the arrival-feasible allocation set, and m is the number of non-anchor arms. An integer vector in $\mathcal{N}$ can be implemented by a legal sequence of pulls: process arms in increasing lifetime order and assign each its required number of unoccupied slots in its available suffix. The suffix constraints guarantee enough slots at each step; unused slots can be assigned to the anchor.

Theorem S11 (Sharp integer approximation)**.** *For every finite arrival schedule,*

$$\kappa\mathscr{C} \le \mathscr{C}_{\text{int}} \le \mathscr{C}, \qquad \kappa = \frac{2\,2^{1/4}}{1+\sqrt{2}}. \tag{S17}$$

The constant κ is the largest universal constant for the lower bound. Numerically, $\kappa \approx 0.98517$.

Proof. For integer $N_i \ge 0$ we have $\phi(N_i) = \sqrt{N_i}$, so the upper bound follows by inclusion of feasible sets. For the lower bound, sort the lifetimes increasingly and let n be any feasible fractional vector

in the corresponding order. Its prefix sums $S_r = \sum_{i=1}^{r} n_i$ satisfy $S_r \le H_{(r)}$. Set $S_0 = 0$, draw a single $U \sim \mathrm{Unif}[0,1)$, and define

$$N_i = \lfloor S_i + U \rfloor - \lfloor S_{i-1} + U \rfloor.$$

These are nonnegative integers, and

$$\sum_{i=1}^{r} N_i = \lfloor S_r + U \rfloor \le H_{(r)},$$

because $H_{(r)}$ is an integer and $U < 1$. Therefore $N \in \mathcal{N}$ for every U. Furthermore, $N_i \in \{\lfloor n_i \rfloor, \lceil n_i \rceil\}$ and $\mathbb{E}N_i = n_i$, by the identity $\mathbb{E}\lfloor x + U \rfloor = x$ for $x \ge 0$.

Let ψ be the continuous piecewise-linear interpolant of $\sqrt{x}$ at the nonnegative integers. The preceding rounding rule gives $\mathbb{E}\sqrt{N_i} = \psi(n_i)$. For $0 \le x \le 1$, $\psi(x) = x = \phi(x)$. On $[q, q+1]$, where $q \ge 1$ is an integer,

$$\psi(x) = \frac{x + \sqrt{q(q+1)}}{\sqrt{q} + \sqrt{q+1}}.$$

Differentiating $\psi(x)/\sqrt{x}$ shows that its minimum on this interval occurs at $x = \sqrt{q(q+1)}$ and equals

$$\kappa_q = \frac{2[q(q+1)]^{1/4}}{\sqrt{q} + \sqrt{q+1}}.$$

To compare these values, put $v_q = \sqrt{(q+1)/q}$. Then $\kappa_q = 2\sqrt{v_q}/(1+v_q)$; the displayed function decreases on $[1, \infty)$, while v_q decreases with q. Hence $\kappa_q \ge \kappa_1 = \kappa$, and $\psi(x) \ge \kappa\phi(x)$ for every $x \ge 0$.

It follows that

$$\mathbb{E}\sum_i \sqrt{N_i} = \sum_i \psi(n_i) \ge \kappa \sum_i \phi(n_i).$$

Apply the rounding to a maximizing fractional allocation. Since every rounded vector is feasible, the integer maximum is at least the displayed expectation. This proves the lower bound, including the empty-arm case.

Finally, let m arms share the lifetime $H_m = \lfloor \sqrt{2}\, m \rfloor$, taking the horizon to be H_m. The only effective allocation constraint is total mass at most H_m. Concavity gives $\mathscr{C} = m\phi(H_m/m) = m\sqrt{H_m/m}$. For integer allocations, an optimum uses all H_m slots and balances the counts to differ by at most one: moving a unit from a larger count to one smaller by at least two increases the square-root sum. Thus, for all sufficiently large m, all counts are 1 or 2, and $\mathscr{C}_{\mathrm{int}} = m\psi(H_m/m)$. Consequently,

$$\frac{\mathscr{C}_{\mathrm{int}}}{\mathscr{C}} = \frac{\psi(H_m/m)}{\sqrt{H_m/m}} \longrightarrow \frac{\psi(\sqrt{2})}{2^{1/4}} = \kappa.$$

No larger universal lower-bound constant is possible. □

The integer allocation result concerns a schedule functional, not a bandit allocation rule. The maximizing allocation depends only on arrival times and does not identify which arms have large means.

S6 Why deterministic-schedule lower bounds cannot be averaged

In an exogenous random-schedule model, the reward laws may be required to be fixed before the schedule is drawn. This is a different decision problem from taking a separate worst-case environment for each realized schedule. The distinction can change the order of the minimax risk even when there are only two arms.

Lemma S12 (Two-arm bound with a known horizon)**.** *For two stationary arms with independent rewards in* $[0,1]$*, there is a policy using the known horizon* H *but not the means such that, if their mean gap is* $\Delta > 0$*,*

$$\mathbb{E}R_H \le \min\left\{H\Delta, \frac{50[1+\log_+(H\Delta^2)]}{\Delta}\right\}. \tag{S18}$$

For $\Delta = 0$ *its regret is zero. The constant is uniform over both reward distributions and every integer* $H \ge 1$*.*

Proof. Use the fixed-horizon MOSS index: an unobserved arm has index $+\infty$, and an arm with $n \ge 1$ observations has index

$$\widehat{\mu}_{i,n} + b(n), \qquad b(n) = \sqrt{\frac{\log_+(H/(2n))}{n}}.$$

All statistics in this lemma start at zero. Fix a deterministic rule for ties. Put $v = H/2$, let i^* be the unique better arm, and write

$$Z = \sup_{n\ge 1}[\mu_{i^*} - \widehat{\mu}_{i^*,n} - b(n)]_+.$$

The maximal-underestimation inequality gives

$$\Pr(Z > x) \le \min\{1, 5/(vx^2)\}, \qquad x > 0. \tag{S19}$$

For completeness, its needed version follows by applying the exponential-supermartingale maximal inequality to $n > v$ and to the dyadic intervals $v2^{-(k+1)} < n \le v2^{-k}$, giving

$$\Pr(Z > x) \le e^{-2vx^2} + \sum_{k\ge 0} 2^{-k} e^{-vx^2/2^{k+1}}.$$

With $u_k = vx^2/2^{k+1}$, the inequality $u_k e^{-u_k} \le 2\int_{u_k/2}^{u_k} e^{-u}\,du$ bounds the sum by $4/(vx^2)$; the first term is at most $1/(vx^2)$. Limits give the same bound for $\Pr(Z \ge x)$.

The suboptimal arm is initialized at most once, at regret at most Δ. On the event $Z \ge \Delta/2$, all remaining regret is bounded by $H\Delta$, whose expected contribution is at most $40/\Delta$ by (S19). On the complementary event, choosing an observed suboptimal arm implies that the better arm has already been observed, since an unobserved arm has infinite index. Index comparison then gives

$$(\widehat{\mu}_{i,n} - \mu_i)_+ + b(n) \ge \Delta/2.$$

Thus either its estimation error is at least $\Delta/4$ or its bonus is at least $\Delta/4$. The global pre-pull count n of this arm is never repeated, and Hoeffding's inequality therefore bounds the expected

number of the first type by

$$\sum_{n\geq 1} e^{-n\Delta^2/8} = \frac{1}{e^{\Delta^2/8}-1} \leq \frac{8}{\Delta^2}.$$

This costs at most $8/\Delta$ in regret.

Suppose first that $H\Delta^2 \geq 1$. If $n \geq \Delta^{-2}$ and $b(n) \geq \Delta/4$, then

$$n \leq \frac{16}{\Delta^2}\log\frac{H}{2n} \leq \frac{16}{\Delta^2}\log(H\Delta^2).$$

Including $n < \Delta^{-2}$, the total number of bonus-type counts is at most $[1+16\log(H\Delta^2)]/\Delta^2$. Their regret is bounded by the same numerator divided by Δ. Since $\Delta \leq 1$, the initialization term is at most $1/\Delta$. Adding terms gives $[50+16\log(H\Delta^2)]/\Delta$, which implies (S18) together with $R_H \leq H\Delta$. If $H\Delta^2 < 1$, the trivial bound $H\Delta < 1/\Delta$ already implies the claimed minimum. No concentration argument conditions on a data-dependent sample count. □

Theorem S13 (A logarithmic separation under an independent schedule)**.** *For every integer $m \geq 1$, there is a distribution $\mathcal{Q}_m$ over two-arm arrival schedules at horizon $T_m = 4^m$ with the following properties. The schedule is independent of the reward streams and policy randomization. Its expected deterministic-schedule capacity is*

$$\mathbb{E}_{\tau\sim\mathcal{Q}_m}\mathscr{C}_{T_m}(\tau) = c_m(m+1), \qquad c_m = \left(\sum_{j=0}^{m} 2^{-j}\right)^{-1} \in [1/2, 1]. \tag{S20}$$

Nevertheless, if the reward laws are chosen before the schedule is drawn, the joint minimax risk satisfies

$$\frac{1}{16} \leq \inf_{\pi}\sup_{P_1,P_2} \mathbb{E}^{\pi}_{\tau\sim\mathcal{Q}_m,P_1,P_2} R_{T_m} \leq 380. \tag{S21}$$

The infimum permits policies to know T_m and $\mathcal{Q}_m$ in advance, while observing the realized arrivals causally. The supremum is over independent stationary $[0,1]$ reward streams, with their distributions fixed independently of the realized schedule. Consequently, no universal positive constant can lower-bound this random-schedule minimax risk by the expected deterministic-schedule capacity.

Proof. Arm 1, the anchor, is available from round 1. Draw J with $\Pr(J=j) = c_m 2^{-j}$, $0 \leq j \leq m$, and let arm 2 arrive at round $T_m - 4^J + 1$. Its remaining lifetime is $H = 4^J$. For a single competitor, $\mathscr{C}_{T_m} = \sqrt{H}$, so (S20) follows immediately.

For the upper bound, play the only arm until the competitor arrives. There is no regret during this prefix. At arrival, the known terminal time reveals H; discard the prefix observations and run the policy of Lemma S12 for the final H rounds. This is a causal policy. Conditional on the schedule, the retained reward suffixes are independent and have the original arm distributions, because the arrival draw is independent and the prefix length is fixed by that draw.

The two fixed reward laws have a single fixed gap $\Delta = |\mu_1 - \mu_2|$. If this gap is zero, the risk is zero. Otherwise set $x_j = \Delta 2^j$. Lemma S12 and $c_m \leq 1$ give

$$\mathbb{E}R_{T_m} \leq \sum_{j=0}^{m} \min\left\{x_j, \frac{50[1+2\log_+ x_j]}{x_j}\right\}.$$

The terms with $x_j < 1$ sum to at most 2. If the remaining set is nonempty, its first value is some $u \in [1, 2)$, since $0 < \Delta \le 1$; subsequent values are $u2^r$. Their sum is at most

$$50 \sum_{r \ge 0} \frac{1 + 2(r+1)\log 2}{2^r} = 50(2 + 8\log 2).$$

Thus the full sum is below 380, uniformly over the two reward laws and m, proving the upper bound.

For the lower bound, restrict to Bernoulli rewards. Fix the anchor mean at $1/2$ and give the competitor mean $1/4$ or $3/4$, each with prior probability $1/2$, independently of J. On the event $J = 0$, the competitor arrives only at the final round. Before that choice, the observed histories have exactly the same distribution under the two environments. If the conditional probability of selecting the competitor is q, its final-round regrets in the two environments are $(1/4)q$ and $(1/4)(1-q)$. Their average is $1/8$, regardless of the policy or prefix observations. All other regret is nonnegative. The prior-average joint risk is therefore at least $\Pr(J = 0)/8 = c_m/8 \ge 1/16$. At least one of the two fixed environments has risk at least this average. This proves the lower bound for every policy in the stated decision class. □

In this example, a hard gap for one remaining lifetime need not be hard for another. The distribution of lifetimes gives comparable weight to each scale after multiplication by its deterministic minimax rate. A single fixed gap incurs only a bounded total weighted loss. This gives a concrete failure of interchanging the environment supremum and the schedule expectation; it does not affect the conditional upper bound for an independent random schedule, or the deterministic-schedule minimax characterization.

S7 A positive characterization for fixed batch templates

The logarithmic separation in Section S6 does not preclude a distribution-dependent characterization on a more restricted class. We now fix the identities and relative timing of finitely many batches, while allowing their common time scale to have an arbitrary finite distribution. The constants below depend on the fixed template. In particular, the result is not a characterization uniform in a growing number of arms or batches.

Let $B_1, \dots, B_b$ be disjoint nonempty sets of non-anchor arms. Fix numbers $\alpha_1 > \cdots > \alpha_b = 1$ and a law Q supported on finitely many positive integers. Assume $\alpha_j h$ is an integer for every $h \in \operatorname{supp}(Q)$ and every j. For a known integer endpoint $T \ge \alpha_1 \max \operatorname{supp}(Q)$, draw $H \sim Q$ independently of all reward streams and policy randomization, and set

$$\tau_j(H) = T - \alpha_j H + 1, \qquad i \in B_j. \tag{S22}$$

Only the anchor is available before $\tau_1(H)$. Each arriving arm remains available. Put

$$k_j = 1 + \sum_{r \le j} |B_r|, \qquad K = k_b, \qquad d_j = \begin{cases} \alpha_j - \alpha_{j+1}, & j < b, \\ 1, & j = b. \end{cases}$$

Thus the interval from batch j through the round before batch $j+1$ has exactly $d_j H$ rounds. Each

such length is a positive integer. The reward laws are arbitrary independent stationary distributions on $[0,1]$, and are fixed before H is drawn. The policy knows the template, T, and Q, and observes the arrivals causally. Let $\mathcal{Q}_{T,Q,\boldsymbol{\alpha},\boldsymbol{B}}$ denote the resulting law of complete schedules. This information class is distinct from that used for the anytime MC-MOSS guarantee.

Define the truncated-lifetime complexity

$$W(Q) = \sup_{x \geq 1} \frac{\mathbb{E}_Q \min\{H, x\}}{\sqrt{x}}. \tag{S23}$$

It is specified directly by the arrival law, with no optimization over bandit policies or reward environments.

Theorem S14 (A fixed-template random-arrival characterization)**.** *Under the preceding assumptions,*

$$\frac{W(Q)}{32} \leq \mathfrak{R}_T^{\mathrm{joint}}(\mathcal{Q}_{T,Q,\boldsymbol{\alpha},\boldsymbol{B}}) \leq A_{\boldsymbol{\alpha},\boldsymbol{B}} W(Q), \qquad A_{\boldsymbol{\alpha},\boldsymbol{B}} = \sum_{j=1}^{b} (20k_j + 43)(k_j - 1)\sqrt{d_j}. \tag{S24}$$

The upper bound is attained by the following causal rule. Play the anchor until the first batch arrives; its arrival reveals $H = (T - \tau_1 + 1)/\alpha_1$. At the start of each batch interval, reset all empirical statistics and run fixed-horizon MOSS on the k_j available arms for its known length $d_j H$. This rule uses the endpoint and template, but need not know the mixing probabilities in Q.

The complexity can be computed by a finite maximization:

$$W(Q) = \max_{h \in \mathrm{supp}(Q)} \frac{\mathbb{E}_Q \min\{H, h\}}{\sqrt{h}}. \tag{S25}$$

Moreover, with $V(Q) = \max_{h \in \mathrm{supp}(Q)} \sqrt{h}\, \mathbb{P}_Q(H \geq h)$,

$$V(Q) \leq W(Q) \leq 2V(Q). \tag{S26}$$

The rule in the theorem is an existence construction for a known endpoint and a fixed template. It is not MC-MOSS, and the theorem does not assert that retaining history is unnecessary for the original problem. Here the regret of every interval is charged explicitly; discarding its predecessor's observations merely gives a convenient upper bound. The lower bound permits the policy to use all previous observations and even the entire realized schedule.

We first establish three ingredients. The first is a deliberately loose fixed-arm consequence of the MOSS index of Audibert and Bubeck (2009). We prove the exact bound used here rather than infer it from a distribution-free statement.

Lemma S15 (A per-arm fixed-horizon bound)**.** *Run fixed-horizon MOSS for $n \geq 1$ rounds on $k \geq 2$ independent stationary $[0,1]$ arms, with all statistics initially zero. An unobserved arm has index $+\infty$; after $r \geq 1$ observations its index is*

$$\widehat{\mu}_{i,r} + \sqrt{\frac{\log_+(n/(kr))}{r}}.$$

Use any fixed deterministic tie rule. If arm i has gap $\Delta_i = \mu^* - \mu_i > 0$, its expected regret contribution

satisfies

$$\Delta_i \mathbb{E}N_i(n) \le \min\left\{ n\Delta_i, \frac{20k + 10 + 16\log_+(n\Delta_i^2)}{\Delta_i} \right\}. \tag{S27}$$

An optimal arm has zero regret contribution. The statement remains valid when $n < k$, when some arms are never initialized, and when the optimal arm is not unique.

Proof. Fix one optimal arm and put $v = n/k$. For its potential sample stream, let

$$Z = \sup_{r \ge 1} \left[\mu^* - \widehat{\mu}_{*,r} - \sqrt{\log_+(v/r)/r} \right]_+ .$$

Lemma S1 yields $\mathbb{P}(Z \ge x) \le \min\{1, 5/(vx^2)\}$ for $x > 0$. Write $\Delta = \Delta_i$. On $\{Z \ge \Delta/2\}$, bounding the contribution of arm i by $n\Delta$ gives at most $20k/\Delta$ in expectation. Its initial pull costs at most $\Delta \le 1/\Delta$. On the complementary event, choosing an observed suboptimal arm implies that the fixed optimal arm has been observed, since otherwise an unobserved arm would have infinite index. Index comparison gives

$$(\widehat{\mu}_{i,r} - \mu_i)_+ + \sqrt{\log_+(v/r)/r} \ge \Delta/2.$$

Either the positive error or the bonus is at least $\Delta/4$. The pre-pull count r is distinct at every selection of arm i. Hoeffding's inequality therefore bounds the expected number of error-type selections by

$$\sum_{r \ge 1} e^{-r\Delta^2/8} = \frac{1}{e^{\Delta^2/8} - 1} \le \frac{8}{\Delta^2}.$$

This argument uses all deterministic potential prefixes and does not condition a concentration bound on an adaptively selected count.

Suppose $n\Delta^2 \ge 1$. Counts below Δ^{-2} contribute at most Δ^{-2} possibilities. If $r \ge \Delta^{-2}$ and the bonus is at least $\Delta/4$, then

$$r \le \frac{16}{\Delta^2} \log \frac{n}{kr} \le \frac{16}{\Delta^2} \log(n\Delta^2).$$

Thus the number of bonus-type counts is at most $[1 + 16\log(n\Delta^2)]/\Delta^2$. Adding the underestimation, initialization, error, and bonus terms gives the second expression in (S27). If $n\Delta^2 < 1$, the deterministic bound $n\Delta < 1/\Delta$ gives the same assertion. Combining it with the deterministic bound $n\Delta$ proves the stated minimum. □

Lemma S16 (A uniform averaging inequality)**.** *Let L be any finitely supported positive integer random variable, and put $W_L = \sup_{x \ge 1} \mathbb{E}\min\{L, x\}/\sqrt{x}$. For every $0 < \Delta \le 1$ and every $a, b \ge 0$,*

$$\mathbb{E}\min\left\{ L\Delta, \frac{a + b\log_+(L\Delta^2)}{\Delta} \right\} \le (1 + a + 2b) W_L. \tag{S28}$$

If $L = dH$ for $d > 0$ and both L and H are positive integer valued, then $W_L = \sqrt{d}\, W(Q)$.

Proof. Set $x = \Delta^{-2} \ge 1$. The contribution from $L \le x$ is at most $\mathbb{E}[L\mathbf{1}\{L \le x\}]/\sqrt{x} \le W_L$. For $t \ge 1$,

$$\mathbb{P}(L > t) \le \frac{\mathbb{E}\min\{L, t\}}{t} \le \frac{W_L}{\sqrt{t}}.$$

Consequently, the contribution from $L > x$ is at most

$$\sqrt{x}\,\mathbb{E}[(a + b\log(L/x))\mathbf{1}\{L > x\}] = \sqrt{x}\left\{a\mathbb{P}(L > x) + b\int_x^\infty \frac{\mathbb{P}(L > t)}{t}\,dt\right\}$$
$$\leq (a + 2b)W_L.$$

The integral identity follows from the nonnegative layer-cake formula; finite support ensures finiteness. This proves the first claim. The scaling identity follows from (S25), proved below: evaluating at the support points dh gives exactly $\sqrt{d}$ times the expressions at h. □

Lemma S17 (A common-environment testing bound)**.** *For every $x \in \mathrm{supp}(Q)$ and every causal policy in the stated information class,*

$$\sup_P \mathbb{E}_{H\sim Q,P} R_T \geq \frac{\mathbb{E}_Q \min\{H, x\}}{32\sqrt{x}}. \tag{S29}$$

The inequality also holds if the policy is told the realized value of H before the experiment starts.

Proof. Choose one label $i_0 \in B_b$. In two environments P_- and P_+, all other arms are Bernoulli with mean $1/2$, while arm i_0 is Bernoulli with mean $1/2 - \delta$ and $1/2 + \delta$, respectively, where $\delta = 1/(8\sqrt{x}) \leq 1/8$. Both environments are fixed independently of H. The equally weighted prior on them is also independent of H.

Condition on $H = h$. Before i_0 arrives, every observed reward has the same law under the two environments. This includes all rewards collected in the earlier batch intervals and every action based on them. After its arrival there are exactly h rounds remaining. Put $\ell_h = \min\{h, x\}$, a positive integer, and retain the full history through the first ℓ_h of those rounds. Only observations from i_0 contribute to the likelihood ratio. If N counts its pulls in these ℓ_h rounds, the chain rule for relative entropy gives

$$\mathrm{KL}(\mathbb{P}_{-,h}, \mathbb{P}_{+,h}) = \mathbb{E}_{-,h}N\ \mathrm{kl}(1/2 - \delta, 1/2 + \delta) \leq 16\delta^2\ell_h \leq \frac{1}{4}.$$

Here $\mathbb{P}_{\pm,h}$ include the entire earlier history and the policy's randomization. The policy kernels agree under both environments, and the schedule is the same after conditioning. The Bernoulli inequality follows from

$$\mathrm{kl}(1/2 - \delta, 1/2 + \delta) = 2\delta\log\frac{1 + 2\delta}{1 - 2\delta} \leq \frac{8\delta^2}{1 - 2\delta} \leq 16\delta^2.$$

Pinsker's inequality yields $\mathrm{TV}(\mathbb{P}_{-,h}, \mathbb{P}_{+,h}) \leq 1/\sqrt{8} < 1/2$. Because $0 \leq N \leq \ell_h$, $|\mathbb{E}_{+,h}N - \mathbb{E}_{-,h}N| \leq \ell_h/2$. The regret in the retained interval is δN in P_- and $\delta(\ell_h - N)$ in P_+. Its prior average is therefore at least

$$\frac{\delta}{2}(\ell_h + \mathbb{E}_{-,h}N - \mathbb{E}_{+,h}N) \geq \frac{\delta\ell_h}{4} = \frac{\min\{h, x\}}{32\sqrt{x}}.$$

All other regret is nonnegative. Average over h; one of the two fixed environments has risk at least the prior average. This closes the lower bound without selecting an environment after observing the schedule. If H is supplied at time zero, its conditional law and all pre-arrival observations are still identical under P_- and P_+, so the proof is unchanged. □

Proof of Theorem S14. We first verify the finite maximization independently of the regret argument.

List the support as $h_1 < \cdots < h_M$. Between consecutive support points,

$$\frac{\mathbb{E}\min\{H,x\}}{\sqrt{x}} = \frac{u}{\sqrt{x}} + v\sqrt{x}, \qquad u, v \geq 0.$$

Its derivative has the sign of $vx - u$, so any interior critical point is a minimum. Below h_1 the expression is $\sqrt{x}$; above h_M it is $\mathbb{E}H/\sqrt{x}$. Its maximum is consequently attained at a support point, proving (S25). This also justifies the scaling assertion of Lemma S16.

For the tail comparison, $\mathbb{E}\min\{H,h\} \geq h\mathbb{P}(H \geq h)$ gives $V(Q) \leq W(Q)$. For every $t > 0$, $\mathbb{P}(H > t) \leq \min\{1, V(Q)/\sqrt{t}\}$: between support points use the smallest support point strictly above t, and above the largest point the tail vanishes. Hence, for $x \geq 1$,

$$\mathbb{E}\min\{H,x\} = \int_0^x \mathbb{P}(H > t)\,dt \leq V(Q)\int_0^x t^{-1/2}\,dt = 2V(Q)\sqrt{x}.$$

This proves (S26).

Take a maximizing support point x in Lemma S17 to obtain the lower bound. For the upper bound, use the rule stated in the theorem. Its pre-first-batch regret is zero. Conditional on $H = h$, all batch boundaries and interval lengths are deterministic, and the arriving batch labels reveal the required schedule. In interval j the available arm set and its best mean are fixed, so each gap $\Delta_{i,j} = \max_{r \leq j,\, i' \in B_r \cup \{a\}} \mu_{i'} - \mu_i$ is the same for every value of h. (The maximum simply ranges over the anchor and batches $1, \ldots, j$.) Resetting the statistics makes this interval a fixed-horizon MOSS experiment with length $d_j h$ and k_j arms.

To justify this last step in the presence of adaptive earlier sampling, one may realize rewards as independent variables $X_{i,t} \sim P_i$ for each label and calendar round and reveal only $X_{I_t,t}$. This has exactly the same observed law as independent potential pull streams. Conditional on the complete past at a batch boundary and on H, future calendar variables remain independent and keep their original laws. Thus the bound in Lemma S15 applies conditionally on the past; no old observations are treated as independent fresh data.

Apply Lemma S16 with $L = d_j H$, $a = 20k_j + 10$, and $b = 16$ to each positive gap. Each arm contributes at most $(20k_j + 43)\sqrt{d_j}W(Q)$. At least one arm has zero gap, so there are at most $k_j - 1$ positive-gap terms. Sum over the batch intervals. They partition every round with more than one available arm, so all prefix and later regret has been included. This yields the upper bound in (S24), uniformly over the fixed reward laws. □

Corollary S18 (When the expected deterministic capacity is correct)**.** *For a fixed template as above, write* $D(Q) = \mathbb{E}_Q\sqrt{H}/W(Q)$. *Then* $D(Q) \geq 1$ *and*

$$\frac{D(Q)}{A_{\boldsymbol{\alpha},\boldsymbol{B}}} \leq \frac{\mathbb{E}_Q \mathscr{C}_T(\tau(H))}{\mathfrak{R}_T^{\mathrm{joint}}(\mathcal{Q}_{T,Q,\boldsymbol{\alpha},\boldsymbol{B}})} \leq 32\sqrt{(K-1)\alpha_1}\, D(Q). \tag{S30}$$

Consequently, over any sequence of laws and horizons with this template fixed, the expected deterministic capacity is of the same order as the joint minimax risk if and only if $D(Q)$ *stays bounded.*

Proof. For every $h \geq 1$, a single final-batch candidate has lifetime h, so its feasible allocation gives $\mathscr{C}_T(\tau(h)) \geq \sqrt{h}$. There are $K - 1$ non-anchor arms, all with lifetime at most $\alpha_1 h$. Since $\phi(n) \leq \sqrt{n}$,

the total-allocation constraint and Cauchy–Schwarz give

$$\mathscr{C}_T(\tau(h)) \le \sqrt{(K-1)\alpha_1 h}.$$

Average these bounds and combine them with Theorem S14. Finally, $\min\{H,x\}/\sqrt{x} \le \sqrt{H}$ pointwise, whence $W(Q) \le \mathbb{E}\sqrt{H}$ and $D(Q) \ge 1$. □

For a deterministic scale $H = h_0$, $W(Q) = \sqrt{h_0}$ and $D(Q) = 1$. If the support lies in $[h_0, Lh_0]$ for a fixed L, then $D(Q) \le \sqrt{L}$. At the other extreme, for $\mathbb{P}(H = 4^j) = c_m 2^{-j}$, $0 \le j \le m$,

$$\frac{\mathbb{E}\min\{H, 4^r\}}{2^r} = c_m\{3 - 2^{-r} - 2^{r-m}\}, \qquad 0 \le r \le m.$$

Thus $1 \le W(Q) < 3$, whereas $\mathbb{E}\sqrt{H} = c_m(m+1)$. For a template with noninteger α_j, choose a fixed positive integer q such that every $q\alpha_j$ is an integer and use $H = q4^J$ instead; both $W(Q)$ and $\mathbb{E}\sqrt{H}$ are multiplied by $\sqrt{q}$. Such a q exists for every template admitting a nonempty legal integer support. The same order-one versus logarithmic separation therefore persists for every fixed multi-batch template covered by the theorem, not only for a single arriving candidate.

The tail functional in (S26) is the familiar weak L^1 tail quantity for $\sqrt{H}$; we do not claim a new general probability functional. The result identifies its statistical role in this specific independent-arrival experiment. Its limitations are material: the template is fixed; the endpoint is known; the first arrival reveals the remaining schedule; and the upper constant grows with the number of arms and batches. Random schedules that do not share a revealing common stretch, and rates uniform in growing templates, remain outside the theorem.

References

Jean-Yves Audibert and Sébastien Bubeck. Minimax policies for adversarial and stochastic bandits. In *Proceedings of the 22nd Annual Conference on Learning Theory*, 2009. URL https://www.microsoft.com/en-us/research/wp-content/uploads/2017/01/COLT09_AB.pdf.

Simai He, Jiawei Zhang, and Shuzhong Zhang. Polymatroid optimization, submodularity, and joint replenishment games. *Operations Research*, 60(1):128–137, 2012. doi: 10.1287/opre.1110.1000.

Emilie Kaufmann, Olivier Cappé, and Aurélien Garivier. On the complexity of best-arm identification in multi-armed bandit models. *Journal of Machine Learning Research*, 17(1):1–42, 2016. URL https://jmlr.org/papers/v17/kaufman16a.html.